

Bounded-Time Logistic SGD Requires Gaussian ReLU Kernel Correlation

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Abstract

Feldman, Kamath, and Srebro ask whether a class that Gaussian-initialized logistic SGD on a ReLU network learns under every input distribution must have dimension complexity at most proportional to the number of parameters times the number of steps. We study their two-layer bias-free process on $\{-1, 1\}^n$, in which both layers are trained by exact single-example updates and the prediction is the sign of the tail sum of scores. Write $B = \eta m T$ for the effective time. For $B \leq 12$ and width and step count large relative to n and B , we show that the expected error on a target h under a marginal D is at least $1/2 - \Delta - 9B A_D(h)$, where $A_D(h)$ is the label correlation of the Gaussian ReLU features under D and Δ is an explicit error term. Once width and step count are also large relative to $1/(1/2 - \varepsilon)$, learning to error $\varepsilon < 1/2$ under every marginal therefore forces correlation at least $(1/2 - \varepsilon)/(18B)$ under every marginal. Convex separation turns this into a margin for every target in one fixed kernel, which bounds the VC dimension by $O(B^2)$ and gives an exact deterministic representation of dimension $O(n)$. The margin is restrictive: adjacent cube points are within $2/\sqrt{n}$ of each other in this kernel, so once $n > (18B/(1/2 - \varepsilon))^2$ only constant targets can be learned under every marginal, and the dimension bound has content only below that threshold. We also prove a parity lower bound and, in a separate small-step regime, an exact fixed-gate representation with a probabilistic-dimension bound. The proofs of the risk bound, the margin theorem, and the fixed-gate theorem are formalized in Lean 4, apart from the final random-projection step.

Keywords: Stochastic gradient descent; dimension complexity; kernel margin; Gaussian initialization; ReLU networks.

1. Introduction

Feldman et al. (2026) ask whether learning under every input distribution by a standard neural training procedure forces a small exact linear representation. Their Open Question 1 fixes the procedure. A fully connected ReLU network with standard Gaussian initialization is trained by single-example logistic SGD, and the prediction is the sign of the sum of the scores over the second half of training. If this procedure learns every target in a class $\mathcal{H}$ to expected error $\varepsilon < 1/4$ under every input distribution, is the dimension complexity $\text{dc}(\mathcal{H})$ at most $C \cdot ST$, where S is the number of parameters and T the number of steps? We study the two-layer, bias-free case on $\mathcal{X}_n = \{-1, 1\}^n$.

Our main result says that, at bounded effective time, the process can beat random guessing on a target and marginal only through the label correlation of one fixed Gaussian ReLU feature map. Let m be the width, η the step size, and $B = \eta m T$. For $w \sim N(0, I_n/n)$ let $\Phi(x) = \text{ReLU}(\langle w, x \rangle)$, viewed as an element of $\mathcal{K} = L^2(N(0, I_n/n))$, and put $A_D(h) = \|\mathbb{E}_{x \sim D} h(x) \Phi(x)\|_{\mathcal{K}}$. Theorem 1 shows that if $B \leq 12$, $m \geq 576n$, and $T \geq 2B$, then

$$\mathcal{E}(h, D) \geq \frac{1}{2} - \Delta - 9B A_D(h) \tag{1}$$

for every Boolean target h and every marginal D , where $\mathcal{E}(h, D)$ is the expected error of the process and Δ is explicit and tends to zero as m/n and T grow at fixed B . The bound makes no assumption on the margins of the learned scores.

Under every marginal the bound becomes a representation theorem. Let $\theta = 1/2 - \varepsilon$. If the process learns $\mathcal{H}$ to error ε under every marginal and $\Delta \leq \theta/2$, which holds once $m \geq 2^{17}n/\theta$ and $T \geq (216B/\theta)^2$, then $A_D(h) \geq \theta/(18B)$ for all $h \in \mathcal{H}$ and all D . The vectors $\mathbb{E}_D h(x)\Phi(x)$ fill the convex hull of $\{h(x)\Phi(x) : x \in \mathcal{X}_n\}$, so the point of that hull nearest the origin gives a unit vector u_h with $h(x)\langle u_h, \Phi(x) \rangle \geq \theta/(18B)$ at every cube point (Theorem 3). Counting with this margin and applying a random Gaussian projection, as in the classical conversion from margin to dimension (Ben-David et al., 2002; Kamath et al., 2020), then give $\text{dc}(\mathcal{H}) \leq 4 \cdot 10^6 (B/\theta)^4 (n+1)$. Since $S = m(n+1)$, this is the bound that Open Question 1 asks for, with a deterministic feature map that represents every target exactly.

The regime is restrictive, and a two-point marginal shows how much. A target that is not constant takes different values at two adjacent points x and x' of the cube. Since ReLU is 1-Lipschitz, $\|\Phi(x) - \Phi(x')\|_{\mathcal{K}} \leq 2/\sqrt{n}$, so the uniform distribution D on $\{x, x'\}$ has $A_D(h) \leq 1/\sqrt{n}$, and Theorem 1 gives $\mathcal{E}(h, D) \geq 1/2 - \Delta - 9B/\sqrt{n}$ (Corollary 4). If the process learns $\mathcal{H}$ to error ε under every marginal and $\Delta \leq \theta/2$, every target in $\mathcal{H}$ is therefore constant once $n > (18B/\theta)^2$, which at $\varepsilon = 1/4$ and $B = 12$ means $n > 746,496$. Since $\text{dc}(\mathcal{H}) \leq 2^n$ for every class, this alone gives a dimension bound that depends only on B/θ , so in this regime Open Question 1 holds for a simple reason, though with an enormous constant. Theorem 3 is best read as a description of how little the process can learn distribution-free at bounded effective time.

What Theorem 3 adds is quantitative and structural. Every target has margin $\theta/(18B)$ in one fixed kernel. This bounds the VC dimension by $162B^2/\theta^2$, which is $2592B^2$ at $\varepsilon = 1/4$ and, since $B \leq 12$, a constant independent of n , and it gives a dimension bound linear in n , which at $\varepsilon = 1/4$ and $B = 12$ is below 2^n once n exceeds about 50. The linear bound does not follow from the VC bound alone. There are classes on N points with VC dimension 2 and dimension complexity $\Omega(N^{1/2}/\log N)$ (Alon et al., 2016, Theorem 2), so on the cube a class of bounded VC dimension can have dimension complexity far above n ; it is the uniform kernel margin that yields the linear bound.

The proof compares the trained network with simpler recurrences. Writing $b = \sqrt{m}a$ makes the initial output coefficients standard Gaussian and gives output-layer steps of size $\gamma = \eta m$ against normalized ReLU features, so $B = \gamma T$ measures the accumulated functional step. The condition on T keeps individual stochastic steps small at fixed B . A frozen-feature approximation of the scores usually yields a classification result only after small output margins are controlled. Here that control comes from the initialization. Remove the label-dependent part of the population logistic update, keeping its damping term and the Gaussian output initialization. The resulting recurrence has a symmetric score distribution. At the start of the tail its density is at most e times a Gaussian density, and from there the tail score is strictly increasing along a suitable direction, so a small perturbation of the score changes its sign only with small probability. The labels enter the population update through one vector, their correlation μ with the normalized initial features. If μ is small, the trained network stays too close in risk to the symmetric recurrence to beat random guessing by much, and averaging over the hidden layer turns $\|\mu\|$ into $A_D(h)$.

Gates need not stay fixed in this regime. At width proportional to n , training on a point mass changes an unbounded number of gates at that point in the first step, with probability tending to one as $n \rightarrow \infty$ (Proposition 5; Section 4 discusses the dimensions involved). A different small-step regime, $\eta \leq 1/m$ with $m \geq 16n^2T^2/\delta^2$, keeps every gate at a test point outside an initialization-defined set fixed along every training history. There the tail score is exactly linear in mn gated-coordinate features, which gives the probabilistic-dimension bound of Section 5. That regime allows effective time up to T but yields only an approximate, randomized representation.

The proofs of Theorems 1, 3, and 7 are formalized in Lean 4 with Mathlib, starting from a definition of the exact process. Section 6 describes the formal statements and what remains on paper.

1.1. Related work

Kernel-regime analyses. The relation between small parameter movement and fixed-feature behavior is central to lazy-training theory (Chizat et al., 2019), and frozen-gate models that keep products of trained matrices appear in Allen-Zhu et al. (2019, Section 6.2). Our score comparison belongs to this line. It uses the Lipschitz property of ReLU across gate boundaries and does not discard the points at which gates change. Ji and Telgarsky (2020) analyze logistic gradient descent and SGD with gated-coordinate features and Gaussian slab estimates; their output weights are fixed signs and they assume separability in the tangent feature space. Daniely (2017) proves conjugate-kernel learning guarantees with resources that depend on a comparator. These works derive learning from a kernel assumption. We argue in the opposite direction: the premise is the classification error that the process achieves, and the kernel margin is the conclusion.

Distribution dependence. Malach et al. (2021) relate gradient-descent learnability under a fixed input distribution to approximation by simpler classes. Their population-gradient theorem assumes a local regularity condition at initialization, and their Remarks 8 and 10 explain that it does not directly cover single-example SGD or ReLU. Yehudai and Shamir (2020) show that gradient methods can fail on a single ReLU neuron under some bounded input distributions, so success under every marginal, the premise here, is a strong requirement. Theorem 3 and Corollary 4 make precise how strong it is at bounded effective time.

Random features and statistical queries. Karchmer and Malach (2025) state an implication from distribution-free gradient descent to random features for a much broader class of models. Their Theorem 3.3 concerns any differentiable model with p parameters and outputs in $[-1, 1]$, trained for T steps by mini-batch SGD on the squared loss, with gradients clipped and rounded to a grid of width c and a batch size b satisfying $bc^2 \geq \Omega(\log(Tp/\delta))$ for a small constant $\delta > 0$. If such a method reaches squared-loss error at most $1/10$ for every source distribution, the theorem asserts that for every prior over targets, most targets can be approximated under every marginal by a linear combination of $\text{poly}(Tp/c^2)$ random features drawn from one law. Even as stated, this conclusion is weaker in kind than Theorem 3. It bounds an average probabilistic dimension, in which the features are random and a fraction of the targets may be approximated poorly or not at all, while Theorem 3 gives one deterministic map that represents every target exactly. The precision

and batch-size condition also excludes the exact single-example process of Open Question 1. Our result, in turn, covers one architecture at bounded effective time, where Corollary 4 confines nonconstant classes to $n \leq (18B/\theta)^2$.

The proof of their Theorem 3.3 passes through statistical queries. The rounded mini-batch process is simulated by an SQ algorithm with Tp queries of tolerance $c/8$ (Karchmer and Malach, 2025, Theorem 3.4, due to Abbe et al. 2021). This bounds the classical SQ dimension of the class, and their Theorem 4.1, which asserts that the average probabilistic dimension is $O(\text{sq}(\mathcal{H})^{24.01})$, is then invoked. Feldman et al. (2026) report, citing a personal communication from Karchmer and Malach, that the claimed upper bounds do not hold because of a flaw in the proof. Consistent with that report, Mausberg (2026) gives an explicit counterexample to Theorem 4.1 under its stated quantifiers: classes of constant classical SQ dimension whose average probabilistic dimension grows without bound. This rules out the intermediate implication on which the proof of Theorem 3.3 relies. It does not disprove the conclusion of Theorem 3.3 about mini-batch SGD. A route that passes only through SQ learnability is closed for a further reason, since distribution-free SQ learnability does not imply small dimension complexity (Patel, 2026; Mausberg, 2026). Our argument works with the dynamics of the process and uses no SQ simulation.

2. The process and the results

Let $\mathcal{X}_n = \{-1, 1\}^n$, and let $\text{sign}(z) = 1$ for $z \geq 0$ and -1 otherwise. For a class $\mathcal{H}$ of functions $\mathcal{X}_n \rightarrow \{-1, 1\}$, ordinary dimension complexity is

$$\text{dc}(\mathcal{H}) = \min \left\{ d : \exists \phi : \mathcal{X}_n \rightarrow \mathbb{R}^d \forall h \in \mathcal{H} \exists v_h \forall x \in \mathcal{X}_n, h(x) \langle v_h, \phi(x) \rangle > 0 \right\}. \quad (2)$$

The strict inequality makes the representation exact.

Fix integers $n, m, T \geq 1$ and $\eta > 0$ before selecting a target h and a marginal D . Independently initialize

$$(W_0)_{ji} \sim N(0, 1/n), \quad (a_0)_j \sim N(0, 1/m), \quad f_t(x) = a_t^\top \text{ReLU}(W_t x). \quad (3)$$

At step $t = 0, \dots, T-1$, draw a fresh $x_t \sim D$, put $y_t = h(x_t)$, and update simultaneously:

$$\begin{aligned} c_t &= \frac{\eta y_t}{1 + \exp(y_t f_t(x_t))}, & v_t &= \mathbf{1}\{W_t x_t > 0\}, \\ a_{t+1} &= a_t + c_t \text{ReLU}(W_t x_t), & W_{t+1} &= W_t + c_t (a_t \odot v_t) x_t^\top. \end{aligned} \quad (4)$$

Both right sides use the old parameters, and a zero preactivation has derivative zero. Put $I_T = \{\lceil T/2 \rceil, \dots, T\}$ and $q = |I_T|$. The returned classifier is $G = \text{sign } F$, where

$$F(x) = \frac{1}{q} \sum_{t \in I_T} f_t(x). \quad (5)$$

Dividing by q does not change the sign of the tail sum. The risk is

$$\mathcal{E}(h, D) = \mathbb{E}_{W_0, a_0, (x_t)_{t < T} \sim D^T} \Pr_{x \sim D} [G(x) \neq h(x)],$$

with an independent test point; we write $\text{err}_D(G, h) = \Pr_{x \sim D}[G(x) \neq h(x)]$. The learning premise at error ε is

$$\forall h \in \mathcal{H} \ \forall D \in \Delta(\mathcal{X}_n), \quad \mathcal{E}(h, D) \leq \varepsilon. \quad (6)$$

The parameters may depend on $\mathcal{H}$, n , and ε , but they are the same for every pair (h, D) .

The proof uses one feature map, fixed before the target and the marginal:

$$\Phi(x)(w) = \text{ReLU}(\langle w, x \rangle), \quad \mathcal{K} = L^2(N(0, I_n/n)). \quad (7)$$

For every cube point, $\|\Phi(x)\|_{\mathcal{K}}^2 = 1/2$. The label correlation of a target under a marginal is

$$A_D(h) = \|\mathbb{E}_{x \sim D} h(x) \Phi(x)\|_{\mathcal{K}}. \quad (8)$$

Theorem 1 (Risk controlled by kernel correlation) *Suppose $m \geq 576n$, $0 < B := \eta m T \leq 12$, and $\gamma := \eta m = B/T \leq 1/2$. Define*

$$\Delta = \frac{\rho_W}{2} + \rho_a + 24\kappa + \frac{27B}{\sqrt{T}}, \quad \rho_W = \min\{1, 2^{n+1}e^{-m/16384}\}, \quad \rho_a = e^{-m/2}, \quad \kappa = \frac{544n}{m}. \quad (9)$$

For every Boolean target h and every marginal D ,

$$\mathcal{E}(h, D) \geq \frac{1}{2} - \Delta - 9BA_D(h). \quad (10)$$

Point masses are allowed as marginals. The proof also bounds $\Pr_x[F(x)h(x) < 0]$, which is at most the error under any rule for breaking ties, including the error of Feldman et al. (2026), who leave $\text{sign } 0$ unspecified. The result therefore does not depend on the convention $\text{sign } 0 = 1$. The proof uses the derivative at a zero preactivation only through $\|a_t \odot v_t\| \leq \|a_t\|$. Applied to every pair in a learned class, Theorem 1 gives Theorem 3 below, and Corollary 2 records a concrete instance of the thresholds.

Corollary 2 (A numerical instance) *If $m \geq 2^{20}n$, $T \geq 2^{24}$, and $0 < B \leq 12$, then $\Delta \leq 471/4096 < 1/8$, and for every h and D ,*

$$\mathcal{E}(h, D) \geq \frac{3}{8} - 9BA_D(h). \quad (11)$$

Theorem 3 (Kernel margin from success under every marginal) *Let $0 \leq \varepsilon < 1/2$ and $\theta = 1/2 - \varepsilon$. Suppose $m \geq 576n$, $0 < B \leq 12$, $\gamma \leq 1/2$, and $\Delta \leq \theta/2$. If (6) holds for the process (3)–(5), then $A_D(h) \geq \theta/(18B)$ for every $h \in \mathcal{H}$ and every marginal D . Consequently each $h \in \mathcal{H}$ has a vector $u_h \in \mathcal{K}$ with*

$$\|u_h\|_{\mathcal{K}} = 1, \quad h(x)\langle u_h, \Phi(x) \rangle_{\mathcal{K}} \geq \frac{\theta}{18B} \quad (x \in \mathcal{X}_n), \quad (12)$$

and

$$\text{VC}(\mathcal{H}) \leq \frac{162B^2}{\theta^2}, \quad \text{dc}(\mathcal{H}) \leq 4 \cdot 10^6 \left(\frac{B}{\theta}\right)^4 (n+1). \quad (13)$$

The conditions $m \geq 2^{17}n/\theta$ and $T \geq (216B/\theta)^2$ imply $m \geq 576n$, $\gamma \leq 1/2$, and $\Delta \leq \theta/2$. The feature map behind the dimension bound is deterministic and may depend on $\mathcal{H}$, but not on the target or the marginal.

In the regime of Corollary 2, any $\varepsilon < 1/4$ gives $\theta > 1/4 > 2\Delta$, so every target has margin at least $1/(72B)$ in $\mathcal{K}$, and Theorem 3 gives $\text{dc}(\mathcal{H}) \leq C \cdot ST$ with $S = m(n+1)$ and a universal constant C . The next corollary limits the premise far more severely.

Corollary 4 (Adjacent points) *Suppose $m \geq 576n$, $0 < B \leq 12$, and $\gamma \leq 1/2$. For every target h that is not constant there is a marginal D supported on two adjacent points of $\mathcal{X}_n$ with*

$$\mathcal{E}(h, D) \geq \frac{1}{2} - \Delta - \frac{9B}{\sqrt{n}}.$$

Consequently, if (6) holds at error $0 \leq \varepsilon < 1/2$ and $\Delta \leq \theta/2$ with $\theta = 1/2 - \varepsilon$, every $h \in \mathcal{H}$ is constant if $n > (18B/\theta)^2$.

3. Proof outline

Condition first on the initial hidden layer. Its normalized ReLU features are

$$\psi(x) = m^{-1/2} \text{ReLU}(W_0 x).$$

With high probability they satisfy $7/16 \leq \|\psi(x)\|^2 \leq 9/16$ at every cube point. On this event and on $\{\|a_0\| \leq 2\}$, the tail score of the trained network is within κ of a linear logistic-SGD recurrence on the features ψ , uniformly over labeled training histories and test points. This estimate allows gates to change. It follows from a bound on the displacement of the input layer and the global Lipschitz property of ReLU.

The linear recurrence has step size $\gamma = \eta m$ and starts at $b_0 = \sqrt{m} a_0 \sim N(0, I_m)$. Conditionally on each initial vector, its stochastic trajectory differs in mean square by at most $\gamma^2 \Lambda T$ from its population recurrence, where $\Lambda = 9/16$. Nonexpansivity of logistic gradient steps gives this estimate without an exponential dependence on T . Because the estimate holds conditionally, it can later be integrated against the anti-concentration bound, at a classification cost of order $B/\sqrt{T}$.

Now use the identity

$$\frac{y}{1 + e^{yz}} = \frac{y}{2} - \frac{1}{2} \tanh(z/2) \quad (y = \pm 1). \quad (14)$$

For $\mu = \mathbb{E}_D h(x) \psi(x)$, the population map is

$$v \mapsto v + \frac{\gamma}{2} \mu - \frac{\gamma}{2} \mathbb{E}_D [\psi(x) \tanh(v^\top \psi(x)/2)].$$

Deleting $\gamma\mu/2$ gives an odd recurrence that does not depend on the labels. At the start of the tail its density is at most e times the standard Gaussian density, and a directional derivative estimate for the remaining tail gives small-ball probability at most $24r$ on $[-r, r]$. Both steps use bounded effective time: the density bound needs $\lceil T/2 \rceil \gamma$ bounded, and the directional estimate needs $B\Lambda/8 < 1$, that is $B < 128/9$. Symmetry makes the risk of this recurrence exactly $1/2$, and restoring the labels moves the tail coefficients by at most $B\|\mu\|/2$.

Combining these estimates proves (10). The initialization average obeys

$$\mathbb{E}_{W_0} \|\mu\|^2 = \mathbb{E}_w (\mathbb{E}_D h(x) \text{ReLU}(\langle w, x \rangle))^2 = A_D(h)^2.$$

Success under every marginal therefore separates the convex hull of $\{h(x)\Phi(x) : x \in \mathcal{X}_n\}$ from the origin. Its nearest point gives (12). Counting with the margin and a finite Gaussian projection then give ordinary dimension. Figure 1 summarizes how gate crossings and signs are handled. The full arguments are in the appendices.

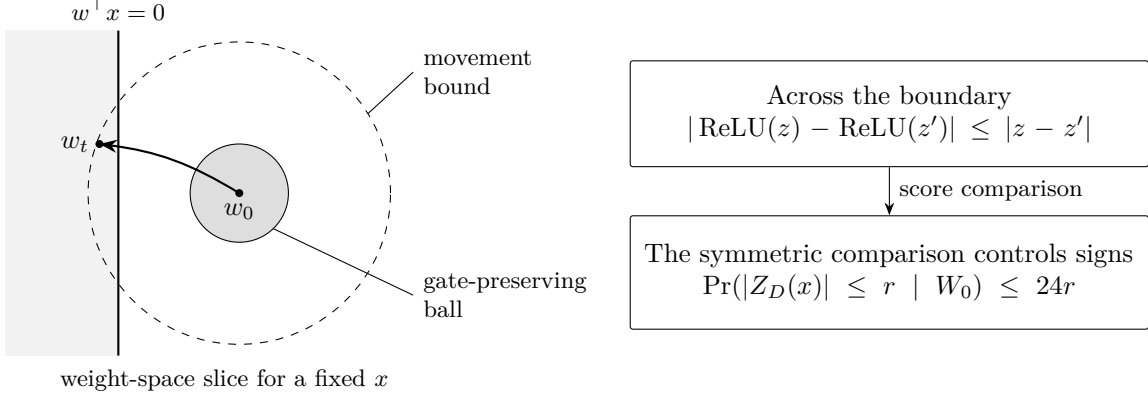


Figure 1: Gate stability gives exact frozen-gate identities, and the score bound used here also controls crossings. Signs are controlled through anti-concentration of a label-symmetric comparison score. In the proof, the movement bound is a Frobenius bound on the displacement of the whole input layer.

4. Gate changes and parity

The regime allows many activation changes at the test point itself, so an argument that discards every point at which some gate changes would not apply here. The next proposition makes this precise at width proportional to n .

Proposition 5 (First-step crossings at linear width) *Fix $C_0 \geq 2^{20}$, an integer $T_0 \geq 2^{24}$, and $B \in (0, 12]$. For each n , set $m_n = \lceil C_0 n \rceil$ and $\eta_n = B/(m_n T_0)$. Choose any $x_n \in \mathcal{X}_n$ and train on its point mass with positive label. Let N_n count the rows for which*

$$w_{0j}^\top x_n > 0 \quad \text{and} \quad w_{1j}^\top x_n < 0.$$

For every fixed integer k_0 , $\Pr[N_n \geq k_0] \rightarrow 1$ as $n \rightarrow \infty$. All these schedules satisfy the conditions of Corollary 2.

The convergence is slow at these constants. For an initially active row a crossing requires $b_j < 0$ and $0 < z_j < \eta_n n |b_j| / \sqrt{m_n}$ in the notation of the proof, so $\mathbb{E}N_n \leq B\sqrt{n}/(2\pi T_0 \sqrt{C_0})$. This is below one for $n < (2\pi T_0 \sqrt{C_0}/B)^2$, about $8 \cdot 10^{19}$ at $B = 12$, and below 10^{-7} at the threshold $n = 746,496$ of Corollary 4. The proposition therefore shows only that the hypotheses of Theorem 1 do not by themselves keep the gates fixed, so that the score comparison has to allow crossings. It concerns the first step and does not show that crossings are common in the dimensions where Theorem 3 allows nonconstant classes.

The kernel correlation also yields a direct obstruction for high-order parities. Under the uniform distribution on the cube, the Gaussian law gives every Fourier coefficient of a fixed degree the same expected squared magnitude, and Parseval's identity bounds their sum by $1/2$.

Corollary 6 (Parity obstruction) *Under the conditions of Corollary 2, for every $U \subseteq [n]$ with $|U| = k$,*

$$\mathcal{E}(\chi_U, \text{Unif}(\mathcal{X}_n)) \geq \frac{3}{8} - \frac{9B}{\sqrt{2^{\binom{n}{k}}}}, \quad \chi_U(x) = \prod_{i \in U} x_i. \quad (15)$$

In particular, if $\binom{n}{k} > 2592B^2$, the process does not learn χ_U to expected error below $1/4$ under the uniform marginal.

5. A fixed-gate regime

A different choice of scales keeps the gates fixed. The conclusion is then a bound on probabilistic dimension in the sense of Kamath et al. (2020); by their Theorem 6, such a bound alone does not bound ordinary dimension. For $\alpha \geq 0$, $\text{pdc}_\alpha(\mathcal{H})$ is the least d for which one law Q of feature maps $\phi : \mathcal{X}_n \rightarrow \mathbb{R}^d$ satisfies

$$\mathbb{E}_{\phi \sim Q} \inf_{u \in \mathbb{R}^d} \Pr_{x \sim D} [\text{sign}\langle u, \phi(x) \rangle \neq h(x)] \leq \alpha \quad \text{for all } h \in \mathcal{H} \text{ and } D \in \Delta(\mathcal{X}_n).$$

The law precedes h and D , while u may depend on them and on the realized map. Use the initial gated coordinates

$$\phi_{W_0}(x)_{j,i} = x_i \mathbf{1}\{\langle w_{0j}, x \rangle > 0\}, \quad 1 \leq j \leq m, \ 1 \leq i \leq n. \quad (16)$$

For $\tau = \eta T \sqrt{n}$ and each initialization, define the exceptional set

$$\mathfrak{B} = \bigcup_{j=1}^m \{x \in \mathcal{X}_n : |\langle w_{0j}, x \rangle| \leq \sqrt{n} r_j\}, \quad r_j = |a_{0j}| \sinh \tau + \|w_{0j}\| (\cosh \tau - 1), \quad (17)$$

and the bound

$$\beta_{n,m}(\tau) = \min \left\{ 1, \frac{(2/\pi) \sqrt{nm} \sinh \tau + \sqrt{2/\pi} m \sqrt{n-1} (\cosh \tau - 1)}{2 - \cosh \tau} \right\}.$$

Theorem 7 (Fixed gates at small total step) *Let $\eta \geq 0$ and $\tau = \eta T \sqrt{n} \leq 1$.*

1. *For every labeled history with labels in $[-1, 1]$ and every $x \notin \mathfrak{B}$, every gate at x keeps its initial nonzero sign up to time T , and $F(x) = \langle C, \phi_{W_0}(x) \rangle$ with $C_{ji} = q^{-1} \sum_{t \in I_T} a_{tj}(W_t)_{ji}$.*
2. *For every fixed marginal D , $\mathbb{E}_{W_0, a_0} D(\mathfrak{B}) \leq \beta_{n,m}(\tau) \leq 2\sqrt{nm} \tau + m\sqrt{n} \tau^2$.*
3. *If (6) holds at error ε , the law of ϕ_{W_0} witnesses $\text{pdc}_{\varepsilon + \beta_{n,m}(\tau)}(\mathcal{H}) \leq mn$. In particular, $\eta \leq 1/m$ and $m \geq 16n^2 T^2 / \delta^2$ with $0 < \delta < 1$ give $\text{pdc}_{\varepsilon + \delta}(\mathcal{H}) \leq mn$.*

The set $\mathfrak{B}$ depends only on the initialization and on (n, m, T, η) , and part 1 holds for all training histories at once, including histories chosen after seeing the initialization. Only the mass estimate needs the marginal to be fixed in advance. Outside $\mathfrak{B}$ the identity in part 1 is exact and keeps every product of trained weights in C , so no output margin is needed to transfer classification error. For $0 < \varepsilon < 1$, taking $\delta = \varepsilon$ gives $\text{pdc}_{2\varepsilon}(\mathcal{H}) \leq mn < S$, an instance of the probabilistic relaxation proposed by Feldman et al. (2026, Section 4). In this regime $B = \eta m T$ may be as large as T , but the width must grow like $n^2 T^2$; Theorems 3 and 7 therefore cover different parts of the parameter space, and neither implies the other.

6. Formal verification

We formalized the proofs of Theorems 1, 3, and 7 in Lean 4 (de Moura and Ullrich, 2021) with Mathlib (The mathlib Community, 2020). The development begins from definitions of the objects in Section 2. The network and the simultaneous update follow (4), with derivative zero at a zero preactivation, and the classifier is the sign of the normalized tail score with $\text{sign } 0 = 1$. Each hidden row has law $N(0, I_n/n)$ and the output layer has law $N(0, I_m/m)$, both obtained from Mathlib’s standard Gaussian measure. A marginal is a probability vector on the cube, so every expectation over training samples or test points is a finite sum, and the risk is the Lebesgue integral over the initialization of the error averaged over histories drawn from D^T . For an integrand not known to be measurable, Mathlib’s Lebesgue integral is the lower integral. A lower bound on it implies the same bound for any measurable version, and an upper bound on a measurable version implies the same bound for it, so the formal lower bounds on the risk are at least as strong as their measurable counterparts and the formal premises are no stronger. No measurability hypothesis on the conditional error is needed. The upper bounds on integrals in parts 2 and 3 of Theorem 7 come with proofs that their integrands are measurable. Table 1 lists the formal counterparts of the paper’s statements. All of them compile without `sorry`, and `#print axioms` reports only `propext`, `Classical.choice`, and `Quot.sound` for each.

Three parts of the paper are not formalized. The first is the dimension bound in Lemma 13 and Theorem 3: the Gaussian projection step, with its chi-square tail bound, is used on paper only, so the formal statements stop at the margin, the VC bound, and the Sauer–Shelah count. The second is Proposition 5. The third is the bound on $\Pr_x[F(x)h(x) < 0]$ stated after Theorem 1. Several formal proofs take a different route from the text. Nonexpansivity of the logistic maps and the Jacobian estimates (31) and (32) are proved with secant slopes of $\tanh$ and of the logistic function, obtained from the one-variable mean value theorem, without Jacobians of the vector-valued maps. Lemma 9 bounds the movement of both layers by induction on t in place of the bootstrap over $\bar{a}$ and D_W , and Lemma 14 uses the addition formulas for $\cosh$ and $\sinh$ in place of the expansion of the matrix power. The first part of Lemma 8 evaluates the two Chernoff exponents at $\zeta = \pm 1/512$ directly instead of bounding $(\log \mathcal{M})''$. The VC bound chooses the signs one at a time with the parallelogram law instead of averaging over random signs. Mathlib does not provide the density of the standard Gaussian measure on $\mathbb{R}^m$, so the formalization identifies it through characteristic functions before applying the change of variables formula in Lemma 11.

Statement	Lean theorems (namespace <code>GaussianSGD</code>)
Lemma 8	<code>init_hidden, init_output</code>
Lemma 9	<code>score_comparison, tail_score_comparison</code>
Lemma 10	<code>frozen_sq_error, label_effect, tail_sq_error,</code> <code>tail_label_effect</code>
Lemma 11	<code>density_bound</code>
Lemma 12	<code>smallball, null_risk</code>
Identity (33)	<code>kernel_identity, kernelCorr_eq_norm, Phi_norm_sq</code>
Theorem 1	<code>risk_lower_bound</code>
Corollary 2	<code>lossTerm_regime, risk_lower_bound_regime</code>
Lemma 13, margin and growth	<code>margin_and_vc, growth_bound</code>
Theorem 3, margin and VC	<code>kernel_margin, lossTerm_le_half, regime_of_large</code>
Margin $1/(72B)$ for $\varepsilon < 1/4$	<code>kernel_margin_regime</code>
Corollary 4	<code>edge_lower_bound, class_constant</code>
Corollary 6	<code>parity_lower_bound, kernelCorr_parity_sq_le</code>
Lemma 14	<code>row_movement</code>
Theorem 7	<code>gates_fixed, exact_representation, gateSet_mass,</code> <code>gateBound_le, measurable_gate_inf, gate_pdc,</code> <code>gate_small_step</code>

Table 1: Formal counterparts of the paper’s statements. The file `AxiomAudit.lean` prints the axioms of each.

7. Scope

By Corollary 4, the premise of Theorem 3 admits nonconstant classes only for $n \leq (18B/\theta)^2$. The main theorems hold at bounded effective time. The bound $B \leq 12$ enters through the density and anti-concentration lemmas and also fixes the constants of the score comparison; we do not know whether success at larger effective time, for instance with B growing with n , forces any representation bound. The thresholds on m and T are sufficient conditions chosen to keep the proofs short and do not locate a transition, and the constants are large: at $\varepsilon = 1/4$ and $B = 12$ the dimension bound is of order $10^{13}(n + 1)$. The score comparison allows many gate crossings, but the conclusion is a kernel-regime statement and gives no separation between feature learning and kernel methods. The representation in Theorem 3 is existential, since its last step is a random projection, and we give no efficient construction of the feature map.

AI Disclosure

GPT-6 Astra and Claude models were used to assist with this research. Samuel Mausberg is responsible for all of its content.

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Appendix A. Initialization and the score comparison

Throughout the appendices recall $\Lambda = 9/16$, and let $\lambda = 7/16$ and $\psi(x) = m^{-1/2} \text{ReLU}(W_0 x)$. Define the good event

$$\mathcal{G} = \{W_0 : \lambda \leq \|\psi(x)\|^2 \leq \Lambda \text{ for every } x \in \mathcal{X}_n\}. \quad (18)$$

It is fixed before any target, marginal, or sample history is chosen.

Lemma 8 (Initialization bounds) *The initialization satisfies*

$$\Pr(W_0 \notin \mathcal{G}) \leq \rho_W, \quad \Pr(\|a_0\| > 2) \leq \rho_a.$$

Proof Fix x . The variables $Y_j = \text{ReLU}(w_{0j}^\top x)^2$ are independent with the law of $Y = Z^2 \mathbf{1}\{Z > 0\}$ for $Z \sim N(0, 1)$. Its mean is $1/2$ and

$$\mathcal{M}(\zeta) = \mathbb{E}e^{\zeta Y} = \frac{1 + (1 - 2\zeta)^{-1/2}}{2}.$$

For $|\zeta| \leq 1/4$, $\mathcal{M} \geq 1/2$ and $\mathcal{M}'' \leq (3/2)2^{5/2}$. Hence $(\log \mathcal{M})'' \leq 32$. Since $(\log \mathcal{M})'(0) = 1/2$, integration gives

$$\log \mathbb{E}e^{\zeta(Y-1/2)} \leq 16\zeta^2.$$

Apply Chernoff's inequality with $\zeta = \pm 1/512$. Each tail at distance $1/16$ from the mean has probability at most $e^{-m/16384}$. A union bound over the 2^n points proves the first claim.

Also $m\|a_0\|^2$ has the χ_m^2 law. Markov's inequality applied to $e^{\chi_m^2/4}$ gives

$$\Pr(\|a_0\| > 2) \leq e^{-m} 2^{m/2} \leq e^{-m/2}.$$

■

Put $g(y, z) = y/(1 + e^{yz})$. For $y = \pm 1$ it has absolute value at most one, and its derivative in z lies between $-1/4$ and zero. On any labeled history, define the linear recurrence

$$u_0 = \sqrt{m} a_0, \quad u_{t+1} = u_t + \gamma g(y_t, u_t^\top \psi(x_t)) \psi(x_t), \quad \gamma = \eta m, \quad (19)$$

driven by the same samples as the network.

Lemma 9 (Score comparison through gate changes) *Suppose $W_0 \in \mathcal{G}$, $\|a_0\| \leq 2$, $m \geq 576n$, $B \leq 12$, and $\gamma \leq 1/2$. For every history with labels $y_t = \pm 1$,*

$$\max_{0 \leq t \leq T} \max_{x \in \mathcal{X}_n} |f_t(x) - u_t^\top \psi(x)| \leq \kappa = 544n/m. \quad (20)$$

The same bound holds for the normalized tail scores.

Proof Write

$$\bar{a} = \max_{t \leq T} \|a_t\|, \quad D_W = \max_{t \leq T} \|W_t - W_0\|_F.$$

From the input-layer updates,

$$D_W \leq \eta T \sqrt{n} \bar{a} = \frac{B\sqrt{n}}{m} \bar{a}. \quad (21)$$

ReLU is 1-Lipschitz, so

$$\|\text{ReLU}(W_t x)\| \leq \sqrt{m\Lambda} + \sqrt{n} D_W.$$

Summing the output-layer increments and using (21) yields

$$\bar{a} \leq \|a_0\| + B\sqrt{\Lambda/m} + \frac{B^2 n}{m^2} \bar{a}.$$

Under the hypotheses, $B^2 n/m^2 \leq 1/4$ and $B\sqrt{\Lambda/m} \leq 3/8$. Thus $\bar{a} < 4$ and $D_W \leq 4B\sqrt{n}/m$.

Let $b_t = \sqrt{m} a_t$ and define

$$e_t(x) = m^{-1/2}(\text{ReLU}(W_t x) - \text{ReLU}(W_0 x)), \quad r_t(x) = f_t(x) - b_t^\top \psi(x).$$

Uniformly in t and x ,

$$\|e_t(x)\| \leq e_* := \frac{4Bn}{m^{3/2}}, \quad |r_t(x)| \leq r_* := \frac{16Bn}{m}. \quad (22)$$

For fixed (x, y) , the map $\mathcal{S}_{x,y}(v) = v + \gamma g(y, v^\top \psi(x)) \psi(x)$ has derivative $I - \gamma \ell'' \psi(x) \psi(x)^\top$, where $\ell'' = -\partial_z g(y, z) = e^{yz}/(1 + e^{yz})^2 \in (0, 1/4]$. As $\gamma \leq 1/2$ and $\|\psi(x)\|^2 \leq \Lambda$, this derivative is positive semidefinite with operator norm at most one, so $\mathcal{S}_{x,y}$ is nonexpansive. Comparing the update of b_t with $\mathcal{S}_{x_t, y_t}(b_t)$ gives a residual of norm at most

$$\gamma(\sqrt{\Lambda} r_*/4 + e_*).$$

Iterating nonexpansivity from $b_0 = u_0$ proves

$$\|b_t - u_t\| \leq B(\sqrt{\Lambda} r_*/4 + e_*).$$

Consequently,

$$|f_t(x) - u_t^\top \psi(x)| \leq (1 + B\Lambda/4)r_* + B\sqrt{\Lambda} e_*.$$

The first term is at most $516n/m$. The second is at most $432n/m^{3/2} \leq 18n/m$, since $m \geq 576$. Their sum is at most $534n/m \leq 544n/m$. Averaging over I_T preserves the bound. $\blacksquare$

Appendix B. Population comparison and Gaussian anti-concentration

Fix $W_0 \in \mathcal{G}$, a Boolean target h , and a marginal D . Let

$$\mu = \mathbb{E}_D h(x) \psi(x), \quad R(v) = v - \frac{\gamma}{2} \mathbb{E}_D [\psi(x) \tanh(v^\top \psi(x)/2)].$$

The map R is odd. Its derivative is

$$DR(v) = I - \gamma H(v), \quad H(v) = \frac{1}{4} \mathbb{E}_D [\psi(x) \psi(x)^\top \text{sech}^2(v^\top \psi(x)/2)], \quad (23)$$

and $0 \preceq H(v) \preceq (\Lambda/4)I$. Both R and $R + \gamma\mu/2$ are nonexpansive when $\gamma \leq 1/2$.

With the same initial vector $b_0 = \sqrt{m}a_0$, define the population and symmetric recurrences

$$v_0 = z_0 = b_0, \quad v_{t+1} = R(v_t) + \gamma\mu/2, \quad z_{t+1} = R(z_t). \quad (24)$$

By (14), $v \mapsto R(v) + \gamma\mu/2$ is the conditional mean of one step of (19).

Lemma 10 (Sampling and label effects) *Assume $W_0 \in \mathcal{G}$ and $\gamma \leq 1/2$. Conditionally on W_0 and b_0 , for $t \leq T$,*

$$\mathbb{E}\|u_t - v_t\|^2 \leq t\gamma^2\Lambda. \quad (25)$$

For every b_0 ,

$$\|v_t - z_t\| \leq t\gamma\|\mu\|/2. \quad (26)$$

In particular, if $\bar{u} = q^{-1} \sum_{t \in I_T} u_t$ and similarly for $\bar{v}$ and $\bar{z}$, then

$$\mathbb{E}[\|\bar{u} - \bar{v}\|^2 \mid W_0, b_0] \leq \frac{\Lambda B^2}{T}, \quad |\langle \bar{v} - \bar{z}, \psi(x) \rangle| \leq \frac{3B}{8}\|\mu\| \quad (x \in \mathcal{X}_n). \quad (27)$$

Proof Let $P = R + \gamma\mu/2$. Given the past, the sample update (19) has conditional mean $P(u_t)$. Write it as $P(u_t) + \gamma\xi_t$, where $\mathbb{E}[\xi_t \mid \text{past}] = 0$ and

$$\mathbb{E}[\|\xi_t\|^2 \mid \text{past}] \leq \mathbb{E}_D \|g(h(x), u_t^\top \psi(x))\psi(x)\|^2 \leq \Lambda.$$

The cross term vanishes in the conditional second moment, so nonexpansivity of P gives

$$\mathbb{E}[\|u_{t+1} - v_{t+1}\|^2 \mid \text{past}] \leq \|u_t - v_t\|^2 + \gamma^2\Lambda.$$

Induction proves (25). Nonexpansivity of R gives $\|v_{t+1} - z_{t+1}\| \leq \|v_t - z_t\| + \gamma\|\mu\|/2$, which proves (26). Jensen's inequality for the tail average and $t\gamma^2\Lambda \leq T\gamma^2\Lambda = \Lambda B^2/T$ give the first assertion of (27). The second uses $\|\psi(x)\| \leq \sqrt{\Lambda} = 3/4$ and $t\gamma \leq B$. $\blacksquare$

Lemma 11 (Density at the beginning of the tail) *Assume $\|\psi(x)\|^2 \leq \Lambda$ for all x , $B \leq 12$, and $\gamma \leq 1/2$, and put $s = \lceil T/2 \rceil$. For $b_0 \sim N(0, I_m)$ and every Borel set $E \subseteq \mathbb{R}^m$,*

$$\Pr(z_s \in E) \leq e \Pr(b_0 \in E). \quad (28)$$

Equivalently, z_s has a density bounded by $e(2\pi)^{-m/2} \exp(-\|v\|^2/2)$.

Proof For every v , the matrix $H(v)$ is positive semidefinite with $\text{tr} H(v) \leq \Lambda/4$, since $\text{sech}^2 \leq 1$ and $\mathbb{E}_D \|\psi(x)\|^2 \leq \Lambda$. Let $\omega = \gamma\Lambda/4 \leq 9/128$. The eigenvalues $\nu_1, \dots, \nu_m$ of $\gamma H(v)$ are nonnegative with $\sum_i \nu_i \leq \omega$, so

$$\det DR(v) = \prod_{i=1}^m (1 - \nu_i) \geq 1 - \sum_{i=1}^m \nu_i \geq 1 - \omega.$$

The trace bound keeps this factor independent of the width. Since $\tanh$ is increasing and 1-Lipschitz, $\langle R(v) - R(v'), v - v' \rangle \geq (1 - \omega)\|v - v'\|^2$, so R is injective. Also R is nonexpansive and fixes zero, so $\|R(b)\| \leq \|b\|$ and $\varphi(R(b)) \geq \varphi(b)$ for the standard Gaussian density φ .

Fix a Borel set E and put $E' = R^{-1}(E)$, so that $R(E') \subseteq E$. The change of variables formula for the injective map R gives

$$\Pr(b_0 \in E) \geq \int_{R(E')} \varphi = \int_{E'} \det DR(b) \varphi(R(b)) db \geq (1 - \omega) \Pr(b_0 \in E').$$

Thus the law of $R(b_0)$ is at most $(1 - \omega)^{-1}$ times the standard Gaussian law. Pushing forward by R preserves such a domination, so by induction the law of $z_s = R^s(b_0)$ is at most $(1 - \omega)^{-s}$ times the standard Gaussian law. Finally $(1 - \omega)^{-s} \leq \exp(s\omega/(1 - \omega)) \leq e$, because $s\omega \leq 1 - \omega$: indeed $\gamma(s + 1) \leq (B + \gamma)/2 + \gamma \leq 27/4$, so $\omega(s + 1) = 9\gamma(s + 1)/64 \leq 243/256$. This proves (28). $\blacksquare$

Lemma 12 (Anti-concentration of the symmetric tail) *Assume $W_0 \in \mathcal{G}$, $B \leq 12$, and $\gamma \leq 1/2$. For each $x \in \mathcal{X}_n$, let*

$$Z_D(x; b_0) = \langle \bar{z}(b_0), \psi(x) \rangle, \quad b_0 \sim N(0, I_m).$$

For every $r \geq 0$,

$$\Pr_{b_0}(|Z_D(x; b_0)| \leq r \mid W_0) \leq 24r. \quad (29)$$

The law of $Z_D(x; b_0)$ is symmetric and has no atom at zero. Hence, for every Boolean h ,

$$\mathbb{E}_{b_0} \text{err}_D(\text{sign } Z_D(\cdot; b_0), h) = 1/2. \quad (30)$$

Proof Put $s = \lceil T/2 \rceil$ and $L = T - s$, and regard the tail as a function of $v = z_s$. Write $z_0^{[v]} = v$ and $z_{j+1}^{[v]} = R(z_j^{[v]})$ for $0 \leq j < L$. The Jacobians $J_j = D_v z_j^{[v]}$ obey

$$J_0 = I, \quad J_{j+1} = (I - \gamma H(z_j^{[v]}))J_j.$$

Each factor is a contraction, so $\|J_j\|_{\text{op}} \leq 1$. Summing $J_{j+1} - J_j = -\gamma H(z_j^{[v]})J_j$ gives

$$\|J_j - I\|_{\text{op}} \leq j\gamma\Lambda/4 \leq B\Lambda/8 \leq 27/32. \quad (31)$$

Since $e^\top J e \geq 1 - \|J - I\|_{\text{op}}$ for every unit vector e , this bound suffices below, although the J_j need not be symmetric.

Suppose first that v is standard Gaussian. For fixed x , put $e_x = \psi(x)/\|\psi(x)\|$ and decompose $v = \beta e_x + v_\perp$. Conditionally on $v_\perp$, β is a standard Gaussian. At every β and $v_\perp$,

$$\begin{aligned} \frac{d}{d\beta} \left[\frac{1}{L+1} \sum_{j=0}^L \langle z_j^{[v]}, \psi(x) \rangle \right] &= \|\psi(x)\| e_x^\top \left(\frac{1}{L+1} \sum_{j=0}^L J_j \right) e_x \\ &\geq \sqrt{\lambda} (1 - B\Lambda/8) \geq \frac{5\sqrt{7}}{128}. \end{aligned} \quad (32)$$

The preimage of $[-r, r]$ in β is therefore an interval of length at most $256r/(5\sqrt{7})$, and the maximum of the one-dimensional Gaussian density bounds its probability by

$$\frac{256r}{5\sqrt{14\pi}} < 8r.$$

By Lemma 11 the law of the actual z_s is at most e times the standard Gaussian law, so the actual small-ball probability is at most $8er < 24r$.

The recurrence R is odd, so $Z_D(x; -b_0) = -Z_D(x; b_0)$. The small-ball estimate excludes an atom at zero. Symmetry then gives error $1/2$ at every fixed test point, and averaging over D proves (30). $\blacksquare$

Appendix C. Proofs of the main theorems

Proof [Proof of Theorem 1] Fix h and D before initialization, and condition on $W_0 \in \mathcal{G}$. For a test point x , put

$$Z = Z_D(x; b_0), \quad d_0 = \kappa + \frac{3B}{8}\|\mu\|, \quad E_x = \langle \bar{u} - \bar{v}, \psi(x) \rangle.$$

Lemmas 9 and 10 show that on $\{\|a_0\| \leq 2\}$,

$$|F(x) - Z| \leq d_0 + |E_x|, \quad \mathbb{E}[|E_x|^2 \mid W_0, b_0] \leq \sigma^2 := \frac{\Lambda^2 B^2}{T}.$$

The second bound holds conditionally on every b_0 . This lets us integrate the error probability against the Gaussian initialization with no independence between E_x and Z .

If $\text{sign } F(x) \neq \text{sign } Z$, then $|Z| \leq |F(x) - Z|$. Conditionally on b_0 with $\|a_0\| \leq 2$ and $|Z| > d_0$, Markov's inequality therefore bounds the probability of disagreement by $\min\{1, \sigma^2/(|Z| - d_0)^2\}$. Give this minimum the value one when $|Z| \leq d_0$. By the layer-cake formula and Lemma 12, its expectation over the initialization is at most

$$\begin{aligned} \int_0^1 \Pr_{b_0} \left(|Z| \leq d_0 + \frac{\sigma}{\sqrt{t}} \mid W_0 \right) dt &\leq 24 \int_0^1 \left(d_0 + \frac{\sigma}{\sqrt{t}} \right) dt \\ &= 24d_0 + 48\sigma. \end{aligned}$$

The symmetric classifier has expected error $1/2$. The same bound holds with $\{\text{sign } F(x) \neq \text{sign } Z\}$ replaced by $\{Zh(x) < 0 \leq F(x)h(x)\}$, on which also $|Z| \leq |F(x) - Z|$, and $\Pr(Zh(x) < 0) = 1/2$ by Lemma 12; this gives the bound on $\Pr_x[F(x)h(x) < 0]$ stated after the theorem. Averaging over the test marginal and adding the probability of $\{\|a_0\| > 2\}$ to the disagreement gives

$$\mathbb{E}[\text{err}_D(G, h) \mid W_0] \geq \frac{1}{2} - \rho_a - 24\kappa - \frac{27B}{\sqrt{T}} - 9B\|\mu\|.$$

The event $\{\|a_0\| > 2\}$ is paid for separately because Lemma 12 uses the unconditioned Gaussian law of b_0 .

For initializations outside $\mathcal{G}$, keep only the nonnegativity of the risk. After averaging,

$$\mathcal{E}(h, D) \geq \frac{1}{2}(1 - \rho_W) - \rho_a - 24\kappa - \frac{27B}{\sqrt{T}} - 9B\mathbb{E}[\mathbf{1}_{\mathcal{G}}\|\mu\|].$$

The hidden rows are independent and identically distributed, which gives the exact identity

$$\begin{aligned} \mathbb{E}_{W_0}\|\mu\|^2 &= \frac{1}{m} \sum_{j=1}^m \mathbb{E}_{w_{0j}} \left(\mathbb{E}_D h(x) \text{ReLU}(w_{0j}^\top x) \right)^2 \\ &= A_D(h)^2. \end{aligned} \tag{33}$$

Thus $\mathbb{E}[\mathbf{1}_{\mathcal{G}}\|\mu\|] \leq A_D(h)$ by the Cauchy–Schwarz inequality. This proves (10). $\blacksquare$

Proof [Proof of Corollary 2] The conditions give $m \geq 576n$ and $\gamma = B/T \leq 12/2^{24} < 1/2$, so Theorem 1 applies. Also $\rho_W, \rho_a \leq 1/64$ and

$$\frac{27B}{\sqrt{T}} \leq \frac{324}{4096}, \quad 24\kappa \leq \frac{51}{4096}.$$

With $\rho_W/2 \leq 32/4096$ and $\rho_a \leq 64/4096$, the total loss is at most $471/4096 < 1/8$, and (10) gives (11). $\blacksquare$

Lemma 13 (Convex separation and ordinary dimension) *Let X be a nonempty finite set with M points, let $\Phi : X \rightarrow \mathcal{K}$ satisfy $\|\Phi(x)\|^2 \leq 1/2$, and let $c > 0$. Suppose a nonempty class $\mathcal{H}$ of functions $X \rightarrow \{-1, 1\}$ satisfies*

$$\|\mathbb{E}_{x \sim D} h(x) \Phi(x)\| \geq c \quad \text{for all } h \in \mathcal{H} \text{ and } D \in \Delta(X). \quad (34)$$

Then every $h \in \mathcal{H}$ has a unit vector u_h with $h(x)\langle u_h, \Phi(x) \rangle \geq c$ for all x , and $\text{VC}(\mathcal{H}) \leq 1/(2c^2)$. With $v = \lfloor 1/(2c^2) \rfloor$,

$$|\mathcal{H}| \leq \sum_{i=0}^v \binom{M}{i} \leq (M+1)^v, \quad \text{dc}(\mathcal{H}) \leq 1 + \frac{32(v+2)}{c^2} \log(M+1).$$

Proof For fixed h , the convex hull

$$C_h = \text{conv}\{h(x)\Phi(x) : x \in X\}$$

is compact in a finite-dimensional subspace, and its elements are exactly the vectors $\mathbb{E}_D h(x)\Phi(x)$, one for each marginal D . This is where the hypothesis over all marginals enters: if p_h is the point of C_h nearest to zero, then $\|p_h\| \geq c$. Minimizing the squared norm along the segment from p_h to any $z \in C_h$ gives

$$\langle p_h, z - p_h \rangle \geq 0.$$

Thus $u_h = p_h/\|p_h\|$ has the stated margin at every point. It depends on h but not on D .

Suppose r points are shattered. For every sign vector $s \in \{-1, 1\}^r$, the unit vector of a target realizing s gives

$$rc \leq \left\| \sum_{i=1}^r s_i \Phi(x_i) \right\|.$$

Averaging the square over independent fair signs gives $r^2 c^2 \leq r/2$, so $r \leq 1/(2c^2)$. The Sauer–Shelah lemma then gives the bound on $|\mathcal{H}|$.

It remains to project the finite collection of input and target vectors. All of them lie in the span of $\Phi(X)$, which is finite-dimensional. A matrix J with independent $N(0, 1/d)$ entries satisfies, for each fixed vector z and $0 < \xi \leq 1/2$,

$$\Pr \left(\left| \|Jz\|^2 - \|z\|^2 \right| > \xi \|z\|^2 \right) \leq 2e^{-d\xi^2/8},$$

by the elementary chi-square Chernoff bound. Apply it to $u_h + \Phi(x)$ and $u_h - \Phi(x)$ for all h and x , with $\xi = c/2$, which is at most $2^{-3/2}$ because $c \leq h(x)\langle u_h, \Phi(x) \rangle \leq \|\Phi(x)\| \leq 2^{-1/2}$. Since $4|\mathcal{H}|Me^{-dc^2/32} < 1$ for some integer $d \leq 1 + 32c^{-2} \log(4|\mathcal{H}|M)$, some realization of J preserves all these squared norms to relative error ξ . Polarization then bounds each inner-product error by

$$\frac{\xi}{2}(\|u_h\|^2 + \|\Phi(x)\|^2) \leq \frac{3c}{8} < c,$$

so $J\Phi(x)$ and Ju_h represent every target exactly. Finally $\log(4|\mathcal{H}|M) \leq \log(4M) + v \log(M+1) \leq (v+2) \log(M+1)$. $\blacksquare$

Proof [Proof of Theorem 3] If $\mathcal{H}$ is empty there is nothing to prove, so assume it is not. Fix $h \in \mathcal{H}$ and D . By Theorem 1 and (6), $1/2 - \Delta - 9BA_D(h) \leq \varepsilon$, so $9BA_D(h) \geq \theta - \Delta \geq \theta/2$. Hence (34) holds with $c = \theta/(18B)$, and Lemma 13 with $M = 2^n$ gives (12) and $\text{VC}(\mathcal{H}) \leq 1/(2c^2) = 162B^2/\theta^2$. For the dimension bound put $K = c^{-2} = 324B^2/\theta^2$. Since $c \leq \|\Phi(x)\| = 2^{-1/2}$, $K \geq 2$ and $v+2 \leq K/2 + 2 \leq 3K/2$. With $\log(2^n + 1) \leq (n+1) \log 2$, the lemma gives

$$\text{dc}(\mathcal{H}) \leq 1 + 48 \log 2 \cdot K^2(n+1) \leq 34K^2(n+1) \leq 4 \cdot 10^6 \left(\frac{B}{\theta}\right)^4 (n+1).$$

For the sufficient conditions, suppose $m \geq 2^{17}n/\theta$ and $T \geq (216B/\theta)^2$. Since $\theta \leq 1/2$, $m \geq 2^{18}n \geq 576n$. If $2B \leq 1$ then $T \geq 1 \geq 2B$; otherwise $T \geq (432B)^2 \geq 2B$. So $\gamma = B/T \leq 1/2$. Next, $24\kappa = 13056n/m \leq 13056\theta/2^{17} < \theta/8$. Put $y = 8/\theta \geq 16$. Then $m/16384 \geq ny$, so $\rho_W/2 \leq 2^n e^{-ny} = e^{-n(y-\log 2)} \leq e^{-(y-\log 2)} \leq 1/y = \theta/8$, since $n \geq 1$ and $y - \log 2 \geq \log y$ for $y \geq 16$. Also $\rho_a = e^{-m/2} \leq e^{-y} < \theta/8$, and $27B/\sqrt{T} \leq \theta/8$. Summing, $\Delta \leq \theta/2$. $\blacksquare$

Proof [Proof of Corollary 4] The cube graph, with edges between points that differ in one coordinate, is connected. If h is not constant, some edge $\{x, x'\}$ therefore has $h(x') = -h(x)$; let i be the coordinate in which x and x' differ, and let D be the uniform distribution on $\{x, x'\}$. Then $\mathbb{E}_D h\Phi = \frac{1}{2}h(x)(\Phi(x) - \Phi(x'))$. Since ReLU is 1-Lipschitz and $w \sim N(0, I_n/n)$,

$$\|\Phi(x) - \Phi(x')\|_{\mathcal{K}}^2 \leq \mathbb{E}_w \langle w, x - x' \rangle^2 = 4\mathbb{E}_w w_i^2 = \frac{4}{n},$$

so $A_D(h) \leq 1/\sqrt{n}$, and Theorem 1 gives the first claim. Under these conditions a nonconstant $h \in \mathcal{H}$ would satisfy $\varepsilon \geq \mathcal{E}(h, D) \geq 1/2 - \theta/2 - 9B/\sqrt{n}$, that is $\sqrt{n} \leq 18B/\theta$. $\blacksquare$

Appendix D. Proofs of the crossing and parity statements

Proof [Proof of Proposition 5] Write $z_j = w_{0j}^\top x_n$ and $b_j = \sqrt{m}a_{0j}$. The pairs (z_j, b_j) are independent pairs of independent standard Gaussians. The initial score

$$f_0(x_n) = m^{-1/2} \sum_j b_j \text{ReLU}(z_j)$$

has mean zero and variance $1/2$ for every n and m .

Fix $r_0 > 0$. On $\{f_0(x_n) \leq r_0\}$, an initially active row updates according to

$$w_{1j}^\top x_n = z_j + \frac{\eta_n n b_j}{\sqrt{m}(1 + e^{f_0(x_n)})},$$

where the update uses the old output coefficient, as simultaneous SGD requires. The row becomes strictly negative whenever

$$b_j \leq -1, \quad 0 < z_j < d_n, \quad d_n = \frac{\eta_n n}{\sqrt{m}(1 + e^{r_0})}.$$

Let C_n count these candidate rows. It is binomial, and

$$\mathbb{E}C_n = m \Pr(b_j \leq -1) \Pr(0 < z_j < d_n) \asymp_{r_0} \frac{Bn}{T_0 \sqrt{m}} \rightarrow \infty,$$

because $m = \lceil C_0 n \rceil$, $d_n \rightarrow 0$, and B, C_0, T_0, r_0 are fixed. Therefore $\Pr(C_n < k_0) \rightarrow 0$ for every fixed k_0 . The count C_n and the initial score are dependent, but the union bound and the variance of the score give

$$\Pr(N_n < k_0) \leq \Pr(C_n < k_0) + \Pr(f_0(x_n) > r_0) \leq \Pr(C_n < k_0) + \frac{1}{2r_0^2}.$$

Take the upper limit in n , then let $r_0 \rightarrow \infty$. ■

Proof [Proof of Corollary 6] For each w , let $\hat{r}_w(U) = \mathbb{E}_{x \sim \text{Unif}(\mathcal{X}_n)} \chi_U(x) \text{ReLU}(w^\top x)$. The Gaussian law is invariant under coordinate permutations, so

$$\ell_k = \mathbb{E}_w \hat{r}_w(U)^2$$

depends only on $|U| = k$. Parseval's identity and nonnegativity give

$$\binom{n}{k} \ell_k \leq \sum_{U \subseteq [n]} \mathbb{E}_w \hat{r}_w(U)^2 = \mathbb{E}_w \mathbb{E}_x \text{ReLU}(w^\top x)^2 = \frac{1}{2}.$$

Since $A_{\text{Unif}}(\chi_U)^2 = \ell_k$, substituting in (11) proves (15). The right side exceeds $1/4$ whenever $\binom{n}{k} > 2592B^2$. ■

Appendix E. Proofs for the fixed-gate regime

Write $\nu = \eta\sqrt{n}$. The labels may be any numbers in $[-1, 1]$, so $|c_t| \leq \eta$.

Lemma 14 (Row movement) *For every initialization, every labeled history, and every row j ,*

$$|a_{tj}| \leq |a_{0j}| \cosh(t\nu) + \|w_{0j}\| \sinh(t\nu), \quad \|w_{tj} - w_{0j}\| \leq |a_{0j}| \sinh(t\nu) + \|w_{0j}\| (\cosh(t\nu) - 1).$$

No restriction on η is needed.

Proof Fix a row and write $\mathbf{a}_t = |a_{tj}|$ and $\mathbf{w}_t = \|w_{tj}\|$. Since $|c_t| \leq \eta$ and $\|x_t\| = \sqrt{n}$, the updates give $\mathbf{a}_{t+1} \leq \mathbf{a}_t + \nu \mathbf{w}_t$ and $\mathbf{w}_{t+1} \leq \mathbf{w}_t + \nu \mathbf{a}_t$. The t -th power of $\begin{pmatrix} 1 & \nu \\ \nu & 1 \end{pmatrix}$ has diagonal entries $E_t = \sum_{r \text{ even}} \binom{t}{r} \nu^r$ and off-diagonal entries $O_t = \sum_{r \text{ odd}} \binom{t}{r} \nu^r$, and $\binom{t}{r} \leq t^r / r!$ gives $E_t \leq \cosh(t\nu)$ and $O_t \leq \sinh(t\nu)$. This proves the first bound. For the displacement, sum the increments: $\|w_{tj} - w_{0j}\| \leq \nu \sum_{r < t} \mathbf{a}_r \leq O_t \mathbf{a}_0 + (E_t - 1) \mathbf{w}_0$, using $O_{t+1} = O_t + \nu E_t$ and $E_{t+1} = E_t + \nu O_t$. $\blacksquare$

Proof [Proof of Theorem 7] Put $\sigma_\tau = \sinh \tau$ and $\omega_\tau = \cosh \tau - 1$. If $x \notin \mathfrak{B}$, the Cauchy–Schwarz inequality and Lemma 14 give, for every row and every $t \leq T$,

$$|\langle w_{tj} - w_{0j}, x \rangle| \leq \sqrt{n} r_j < |\langle w_{0j}, x \rangle|,$$

so every preactivation at x keeps its initial nonzero sign along every history. On such a point, $f_t(x) = \sum_j a_{tj} \mathbf{1}\{\langle w_{0j}, x \rangle > 0\} \langle w_{tj}, x \rangle = \sum_{j,i} a_{tj} (W_t)_{ji} \phi_{W_0}(x)_{j,i}$, and averaging over I_T proves part 1.

For part 2 fix x and a row j , and write $w_{0j} = Z/\sqrt{n}$ with $Z \sim N(0, I_n)$. Then $\Gamma = \langle w_{0j}, x \rangle \sim N(0, 1)$ and $\|Z\|^2 = \Gamma^2 + V^2$, where V is independent of Γ with $\mathbb{E}V \leq \sqrt{n-1}$; both are independent of a_{0j} . The radius r_j depends on $\|w_{0j}\|$, and hence on Γ . The j -th event in (17) implies

$$|\Gamma| \leq \sqrt{n} |a_{0j}| \sigma_\tau + \omega_\tau \sqrt{\Gamma^2 + V^2} \leq \sqrt{n} |a_{0j}| \sigma_\tau + \omega_\tau |\Gamma| + \omega_\tau V,$$

and since $\omega_\tau < 1$ for $\tau \leq 1$, it is contained in $\{|\Gamma| \leq (\sqrt{n} |a_{0j}| \sigma_\tau + \omega_\tau V) / (1 - \omega_\tau)\}$, whose right side is independent of Γ . The standard Gaussian density is at most $1/\sqrt{2\pi}$ and $\mathbb{E}|a_{0j}| = \sqrt{2/(\pi m)}$, so this event has probability at most

$$\frac{(2/\pi) \sqrt{n/m} \sigma_\tau + \sqrt{2/\pi} \sqrt{n-1} \omega_\tau}{1 - \omega_\tau}.$$

A union bound over the rows gives $\Pr(x \in \mathfrak{B}) \leq \beta_{n,m}(\tau)$ for each fixed x , and averaging over a fixed marginal gives $\mathbb{E}D(\mathfrak{B}) \leq \beta_{n,m}(\tau)$. The estimate is for a fixed x , which is why part 2 needs the marginal to be fixed in advance. For the simpler bound, $0 \leq \tau \leq 1$ gives $\sinh \tau \leq 6\tau/5$, $\cosh \tau - 1 \leq 5\tau^2/9$, and $2 - \cosh \tau \geq 4/9$; with $2/\pi < 2/3$ and $\sqrt{2/\pi} < 4/5$ this yields $\beta_{n,m}(\tau) \leq 2\sqrt{nm} \tau + m\sqrt{n} \tau^2$.

For part 3 fix h and D . Pathwise, part 1 gives $\text{err}_D(\text{sign}\langle C, \phi_{W_0} \rangle, h) \leq \text{err}_D(G, h) + D(\mathfrak{B})$. The infimum over u of $\text{err}_D(\text{sign}\langle u, \phi_{W_0} \rangle, h)$ depends only on W_0 and is at most the error of every array C , whatever a_0 and the samples. Averaging gives

$$\mathbb{E}_{W_0} \inf_u \text{err}_D(\text{sign}\langle u, \phi_{W_0} \rangle, h) \leq \mathcal{E}(h, D) + \mathbb{E}D(\mathfrak{B}) \leq \varepsilon + \beta_{n,m}(\tau).$$

Finally, if $\eta \leq 1/m$ and $m \geq 16n^2 T^2 / \delta^2$, then $\tau \leq T\sqrt{n}/m \leq \delta^2 / (16n^{3/2} T) < 1$ and $\beta_{n,m}(\tau) \leq 2nT/\sqrt{m} + n^{3/2} T^2 / m \leq \delta/2 + \delta^2 / (16\sqrt{n}) < \delta$. $\blacksquare$